Name: ______________________

Worksheet

Simplifying fractions

Task A: Simplifying fractions using factor pairs

1) Find your way through the maze by simplifying the fractions.
What simplified fraction should go in the end box?

START $\frac{12}{30}$	$\frac{2}{5}$ $\frac{36}{48}$	$\frac{10}{25}$ $\frac{18}{42}$	$\frac{9}{21}$ $\frac{15}{25}$	$\frac{3}{5}$ $\frac{24}{36}$	$\frac{12}{18}$ $\frac{6}{15}$	$\frac{9}{12}$ $\frac{3}{4}$	$\frac{12}{36}$ $\frac{6}{18}$	$\frac{1}{3}$ $\frac{24}{70}$	$\frac{12}{35}$ $\frac{17}{34}$
$\frac{3}{5}$ $\frac{10}{12}$	$\frac{3}{4}$ $\frac{36}{60}$	$\frac{3}{5}$ $\frac{18}{36}$	$\frac{1}{2}$ $\frac{33}{55}$	$\frac{2}{3}$ $\frac{18}{63}$	$\frac{2}{7}$ $\frac{18}{27}$	$\frac{3}{20}$ $\frac{45}{135}$	$\frac{1}{3}$ $\frac{21}{49}$	$\frac{4}{7}$ $\frac{21}{42}$	$\frac{1}{2}$ $\frac{20}{30}$
$\frac{5}{6}$ $\frac{18}{24}$	$\frac{9}{15}$ $\frac{36}{72}$	$\frac{9}{18}$ $\frac{14}{10}$	$\frac{3}{5}$ $\frac{18}{20}$	$\frac{9}{10}$ $\frac{42}{49}$	$\frac{6}{7}$ $\frac{65}{100}$	$\frac{13}{20}$ $\frac{18}{120}$	$\frac{3}{7}$ $\frac{32}{56}$	$\frac{3}{6}$ $\frac{12}{18}$	$\frac{2}{3}$ $\frac{90}{120}$
$\frac{3}{4}$ $\frac{12}{36}$	$\frac{1}{2}$ $\frac{28}{35}$	$\frac{7}{5}$ $\frac{22}{30}$	$\frac{8}{10}$ $\frac{18}{27}$	$\frac{6}{9}$ $\frac{36}{42}$	$\frac{18}{21}$ $\frac{15}{36}$	$\frac{2}{3}$ $\frac{90}{105}$	$\frac{4}{7}$ $\frac{32}{48}$	$\frac{42}{48}$ $\frac{45}{60}$	$\frac{9}{12}$ $\frac{15}{21}$
$\frac{1}{4}$ $\frac{8}{14}$	$\frac{4}{7}$ $\frac{25}{45}$	$\frac{5}{9}$ $\frac{16}{24}$	$\frac{2}{3}$ $\frac{12}{14}$	$\frac{6}{7}$ $\frac{30}{72}$	$\frac{5}{12}$ $\frac{60}{70}$	$\frac{6}{7}$ $\frac{13}{39}$	$\frac{1}{3}$ $\frac{25}{375}$	$\frac{1}{15}$ $\frac{84}{98}$	END

2) Which fraction in each row does not simplify to the fraction in the first column?

a)	$\frac{1}{4}$	$\frac{5}{20}$	$\frac{12}{48}$	$\frac{4}{12}$	$\frac{8}{32}$	$\frac{15}{60}$
b)	$\frac{3}{5}$	$\frac{15}{25}$	$\frac{27}{45}$	$\frac{9}{15}$	$\frac{21}{35}$	$\frac{18}{20}$
c)	$\frac{4}{9}$	$\frac{8}{18}$	$\frac{60}{126}$	$\frac{28}{63}$	$\frac{48}{108}$	$\frac{44}{99}$
d)	$\frac{6}{13}$	$\frac{30}{65}$	$\frac{42}{104}$	$\frac{54}{117}$	$\frac{36}{78}$	$\frac{12}{26}$
e)	$\frac{7}{15}$	$\frac{49}{105}$	$\frac{35}{75}$	$\frac{21}{45}$	$\frac{63}{153}$	$\frac{84}{180}$

Task B: Simplifying fractions using prime factors

Use these prime factorisations to simplify the fractions:

$150 = 2 \times 3 \times 5^2$ $\quad$ $315 = 3^2 \times 5 \times 7$ $\quad$ $360 = 2^3 \times 3^2 \times 5$

$540 = 2^2 \times 3^3 \times 5$ $\quad$ $945 = 3^3 \times 5 \times 7$ $\quad$ $8400 = 2^4 \times 3 \times 5^2 \times 7$

a) $\frac{150}{315}$

b) $\frac{315}{945}$

c) $\frac{150}{8400}$

d) $\frac{315}{360}$

e) $\frac{945}{8400}$

f) $\frac{540}{945}$

g) $\frac{540}{8400}$

h) $\frac{360}{150}$